Strongly cospectral sets in seven-dimensional cubelike graphs

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Abstract

Two vertices of a graph are strongly cospectral if their orthogonal projections onto each adjacency eigenspace agree up to sign. Árnadóttir and Godsil asked in 2021 whether their upper bound $2^{\lceil d/2\rceil-1}$ for the size of a pairwise strongly cospectral set in a cubelike graph on 2^d vertices, $d \geq 3$, is tight in every dimension in its range. We prove that every such set in every cubelike graph on 128 vertices has size at most four, whereas their bound gives eight. Their connected construction attains four, so the maximum in dimension seven is exactly four and the all-dimensions tightness question has a negative answer. The proof combines character sums, congruences and a second-moment bound with an exact finite inequality certified by linear programming duality. The universal upper bound, including the certificate checks, is formalized in Lean; attainment is checked separately.

Computer-assisted; not yet reviewed by experts.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 Introduction

A *cubelike graph* is a Cayley graph $\mathrm{Cay}(\mathbb{F}_2^d, S)$, where $S \subseteq \mathbb{F}_2^d \setminus \{0\}$: vertices x and y are adjacent when $x + y \in S$. For its adjacency matrix A, let E_θ be the orthogonal projection onto the eigenspace with eigenvalue θ, and let e_x be the coordinate vector at x. Vertices x, y are *strongly cospectral* when $E_\theta e_x = \pm E_\theta e_y$ for every eigenvalue θ, with the sign allowed to depend on θ.

Theorem 1.1. *Let $S \subseteq \mathbb{F}_2^7 \setminus \{0\}$. Every set of pairwise strongly cospectral vertices in $\mathrm{Cay}(\mathbb{F}_2^7, S)$ has at most four elements. This bound is attained by a connected graph.*

No connectivity or degree assumption is imposed on the upper bound; in particular, it includes the empty connection set.

Árnadóttir and Godsil [1], Corollary 8.3, proved that a pairwise strongly cospectral set in a cubelike graph on 2^d vertices has size at most $2^{\lceil d/2\rceil-1}$ for $d \geq 3$. Their Example 8.4 attains this bound for $d = 3, 4, 5, 6$. In Section 11 they asked:

> "Is the upper bound on the size of strongly cospectral sets in cubelike graphs tight in all dimensions?"

The question appeared in their first preprint, posted September 15, 2021; the journal article appeared in 2023. At $d = 7$ their bound is eight. Theorem 1.1 reduces it to four, and hence answers their question negatively.

The contribution here is the upper bound four in dimension seven. The character criterion used below is the specialization of Árnadóttir–Godsil's Theorem 10.1 to an elementary abelian 2-group, building on the normal-Cayley character criterion of Peter Sin and Julien Sorci [3, Theorems 2.3 and 2.4], as credited in Árnadóttir–Godsil's Theorem 5.2. The attaining connected example is Árnadóttir–Godsil's Theorem 10.2. Their bound follows from eigenvalue multiplicities and a global second-moment argument. Here the hardest step is a stronger finite inequality, certified by linear programming duality. It combines disjointness of the eigenvalue values on eight character cosets with their individual means, their residues modulo four, and Parseval's identity. The earlier multiplicity bound does not impose these coset restrictions together.

This result does not bound the size of strongly cospectral sets uniformly over all dimensions. Peter Sin [2], in work first posted in 2022, constructed cubelike graphs with arbitrarily large strongly cospectral classes. The maximum in dimensions eight and higher is not determined here. The computational examples and Sage code of Árnadóttir and Godsil, and Gordon Royle's cubelike graph lists credited by them, are prior work.

Section 2 explains the proof. Sections 3 and 4 derive the character and moment restrictions. Section 5 reduces the upper bound to a finite inequality, and Section 6 gives the exact certificate argument. Section 7 describes the known attaining graph. The final section states the scope of the computation and formalization.

2 Proof overview

The proof rules out a strongly cospectral class of size eight. Write $V = \mathbb{F}_2^7$. For a character indexed by $a \in V$, the corresponding adjacency eigenvalue is

$$\lambda_a = \sum_{s \in S} (-1)^{a \cdot s},$$

where the dot product takes values in $\mathbb{F}_2$. The vertices strongly cospectral with zero form a subspace $K(S)$ of V: a vector g lies in it precisely when $a \cdot g$ is constant on each *eigenvalue fiber*, meaning a set $\{a \in V : \lambda_a = \theta\}$. The sum $\sigma = \sum_{s \in S} s$ also belongs to $K(S)$. Consequently, if $|K(S)| \geq 8$, a change of basis places a three-dimensional subspace H inside $K(S)$ with $\sigma \in H$.

The eight cosets of $H^\perp$ then have disjoint sets of eigenvalue values. All values in a given coset have a prescribed residue modulo four. Their full coset means depend only on $B = S \cap H$, which has at most seven elements. After removing the trivial character from the zero coset, its remaining mean also depends on $n = |S|$. The sum of all squared eigenvalues is fixed by Parseval's identity: it is $128|S|$.

For each coset, retain only the nearest eigenvalue values on either side of its mean. These two endpoints give a lower bound for that coset's sum of squares. They must satisfy the residue restriction and must avoid the endpoints chosen by every other coset. The eigenvalue $|S|$ at the trivial character is treated separately. This leaves a finite question about eight pairs of integers, parameterized by $|S|$, B, and σ.

The hard finite inequality is that every permitted selection of these pairs has total lower bound strictly greater than $128|S|$. It is certified by assigning nonnegative penalties to endpoint values. The penalty-adjusted minimum for each coset can be computed independently; subtracting the total available penalty gives a valid lower bound because each endpoint value can be used by at most one coset. Exact certificates make that lower bound exceed $128|S|$ in every case. The resulting contradiction proves the upper bound.

3 The character criterion and normalization

We identify strongly cospectral classes with a subspace defined by character eigenvalues and put a hypothetical large class in a fixed position. Throughout the proof, $V = \mathbb{F}_2^7$, $S \subseteq V \setminus \{0\}$, and $n = |S|$. For $a \in V$, define

$$\chi_a(x) = (-1)^{a \cdot x}, \qquad \lambda_a = \sum_{s \in S} \chi_a(s), \qquad n_a = |\{s \in S : a \cdot s = 1\}|. \tag{1}$$

In particular, $\lambda_a = n - 2n_a$. Define

$$K(S) = \{g \in V : \lambda_a = \lambda_b \Longrightarrow a \cdot g = b \cdot g \text{ for all } a, b \in V\}. \tag{2}$$

Lemma 3.1 (Character criterion)**.** *For $x, y \in V$, the vertices x and y are strongly cospectral in $\mathrm{Cay}(V, S)$ if and only if $x + y \in K(S)$. The set $K(S)$ is a subspace, and its cosets are the strongly cospectral classes.*

Proof. The characters χ_a form an orthogonal basis of $\mathbb{R}^V$, with squared norm 128, and $A\chi_a = \lambda_a \chi_a$. Thus

$$E_\theta e_g = \frac{1}{128} \sum_{a : \lambda_a = \theta} \chi_a(g) \chi_a.$$

By linear independence, this equals $\pm E_\theta e_0$ precisely when $\chi_a(g)$ is constant over a with $\lambda_a = \theta$. This is exactly the condition $g \in K(S)$. Translations are graph automorphisms, so the assertion for x, y follows by translating x to zero. Finally, each condition defining $K(S)$ is a homogeneous linear equation in g, and hence $K(S)$ is a subspace. □

Lemma 3.2. *Put $\sigma = \sum_{s \in S} s$, where the sum is in V. Then $\sigma \in K(S)$, and for every $a \in V$,*

$$n_a \equiv a \cdot \sigma \pmod 2, \qquad \lambda_a \equiv n - 2(a \cdot \sigma) \pmod 4. \tag{3}$$

Here elements of $\mathbb{F}_2$ in integer congruences are represented by $0, 1$.

Proof. The parity of n_a is $\sum_{s \in S} a \cdot s = a \cdot \sigma$. Equality of two eigenvalues implies equality of the corresponding n_a, so it implies equality of the dot products with σ. The second congruence follows from $\lambda_a = n - 2n_a$. □

Let $e_1, \ldots, e_7$ denote the standard basis of V, and put $H = \langle e_1, e_2, e_3 \rangle$. We encode a vector $\sum_i x_i e_i$ by the integer $\sum_i 2^{i-1} x_i$. In particular, $H = \{0, \ldots, 7\}$. Addition of encoded vectors is bitwise exclusive-or, rather than integer addition.

Lemma 3.3 (Normalization)**.** *If $|K(S)| \geq 8$, an invertible linear change of coordinates replaces S by a connection set for which $H \subseteq K(S)$ and $\sigma \in H$.*

Proof. By Lemmas 3.1 and 3.2, $K(S)$ is a subspace of dimension at least three containing σ. Choose a three-dimensional subspace of $K(S)$ containing σ; this is possible also when $\sigma = 0$. Extend a basis of that subspace to a basis of V, and let T map this basis to the standard one. For the connection set TS,

$$\lambda_a^{TS} = \lambda_{T^\intercal a}^{S}, \qquad K(TS) = TK(S), \qquad \sigma(TS) = T\sigma(S).$$

The eigenvalue identity follows from $a \cdot Ts = (T^\intercal a) \cdot s$; substitution in (2) gives the identity for K. These identities give the required position. □

If a pairwise strongly cospectral set has more than four vertices, Lemma 3.1 gives $|K(S)| > 4$. The cardinality of a binary subspace is a power of two, so it is at least eight. It therefore suffices to exclude the normalized situation of Lemma 3.3.

4 Coset means, residues, and the second moment

We now derive the restrictions that cannot hold together in dimension seven. Assume $H \subseteq K(S)$ and $\sigma \in H$. Write $t = \sigma \in \{0, \dots, 7\}$, $B = S \cap H$, and $m = |B|$. For $u \in \{0, \dots, 7\}$, define

$$C_u = \{a \in V : \text{the first three coordinates of } a \text{ equal } u\}, \qquad \mu_u = \sum_{h \in B} (-1)^{u \cdot h}.$$

The sets C_u are the cosets of $H^\perp$; each has 16 elements. In this formula $u \cdot h$ is the three-coordinate binary dot product.

Lemma 4.1. *The following hold:*

1. *If $u \neq w$, no eigenvalue value occurs both on C_u and on C_w.*
2. *For $a \in C_u$, $-n \leq \lambda_a \leq n$ and $\lambda_a \equiv n - 2(u \cdot t) \pmod 4$. Also $\lambda_0 = n$.*
3. *The coset sums and total second moment are*

$$\sum_{a \in C_u} \lambda_a = 16\mu_u, \qquad \sum_{a \in V} \lambda_a^2 = 128n. \tag{4}$$

In particular, $\mu_0 = m$.

Proof. If $u \neq w$, some $e_i \in H$, $1 \leq i \leq 3$, has different dot products with $a \in C_u$ and $b \in C_w$. Since $e_i \in K(S)$, their eigenvalues cannot be equal. The range and trivial eigenvalue follow from (1); the congruence is (3), because $\sigma = t \in H$.

For the coset sum, write $a = (u, v)$ and $s = (h, y)$ according to $V = \mathbb{F}_2^3 \oplus \mathbb{F}_2^4$. Character orthogonality gives

$$\sum_{a \in C_u} \lambda_a = \sum_{(h,y) \in S} (-1)^{u \cdot h} \sum_{v \in \mathbb{F}_2^4} (-1)^{v \cdot y} = 16 \sum_{h \in B} (-1)^{u \cdot h}.$$

Similarly,

$$\sum_{a \in V} \lambda_a^2 = \sum_{s, s' \in S} \sum_{a \in V} (-1)^{a \cdot (s + s')} = 128|S|,$$

since the inner sum is nonzero precisely when $s = s'$. □

For a coset other than C_0, the mean of its 16 eigenvalues is μ_u. On C_0 we separate out $\lambda_0 = n$; the other 15 values have mean $(16m - n)/15$. This distinction retains the contribution of the trivial character exactly.

Lemma 4.2 (Two-point second-moment bound)**.** *Let $x_1, \dots, x_N$ be real numbers with $N \geq 1$ and mean M. Let α be the greatest of these numbers at most M, and let β be the least of them at least M. Then*

$$\sum_{i=1}^{N} x_i^2 \geq N\big(M^2 + (M - \alpha)(\beta - M)\big). \tag{5}$$

Proof. Both endpoints exist by the definition of a mean. No x_i lies strictly between them, so $(x_i - \alpha)(x_i - \beta) \geq 0$. Summing and using $\sum_i x_i = NM$ gives

$$\sum_i x_i^2 \geq N\big((\alpha + \beta)M - \alpha\beta\big) = N\big(M^2 + (M - \alpha)(\beta - M)\big).$$

This includes $\alpha = \beta = M$. □

5 The finite inequality

We state a finite relaxation of the spectral restrictions and show that every hypothetical large class supplies a feasible instance. A *case* is a triple (n, B, t) with

$$0 \le n \le 127, \quad B \subseteq \{1, \dots, 7\}, \quad |B| \le n, \quad t \in \{0, \dots, 7\}. \tag{6}$$

Set $m = |B|$, and, for $u = 0, \dots, 7$, set

$$M_0 = \frac{16m - n}{15}, \qquad M_u = \mu_u = \sum_{h \in B} (-1)^{u \cdot h} \ (u \neq 0), \qquad r_u \equiv n - 2(u \cdot t) \pmod 4. \tag{7}$$

The residue r_u is represented by one of $0, 1, 2, 3$.

Definition 5.1. An *admissible bracket assignment* for a case (n, B, t) consists of eight pairs of integers (α_u, β_u), $0 \le u \le 7$, satisfying:

1. $-n \le \alpha_u \le M_u \le \beta_u \le n$;
2. if $\alpha_u = \beta_u$, then $\alpha_u = M_u$;
3. $\alpha_u \equiv \beta_u \equiv r_u \pmod 4$;
4. if $u \neq 0$, neither endpoint is n;
5. the sets $\{\alpha_u, \beta_u\}$ are pairwise disjoint as u varies.

Its *cost* is $\sum_{u=0}^{7} c_u$, where

$$c_0 = n^2 + (\alpha_0 + \beta_0)(16m - n) - 15\alpha_0\beta_0, \tag{8}$$

$$c_u = 16\big((\alpha_u + \beta_u)\mu_u - \alpha_u\beta_u\big) \quad (u \neq 0). \tag{9}$$

The cost for $u \neq 0$ is the right side of (5) with $N = 16$ and $M = \mu_u$. For $u = 0$ it is n^2 plus that bound with $N = 15$ and $M = M_0$. Although M_0 can be rational, the expanded costs (8)–(9) are integers. A *bracket* means one of these integer pairs. Condition 2 is redundant given condition 1; it is retained to match the finite assertion checked by the accompanying certificates. An endpoint equal to the mean is allowed even when the other endpoint is different.

Lemma 5.2 (Extraction)**.** *If $H \subseteq K(S)$ and $\sigma \in H$, the case $(|S|, S \cap H, \sigma)$ has an admissible bracket assignment of cost at most $128|S|$.*

Proof. For $u \neq 0$, apply Lemma 4.2 to the 16 values on C_u; for $u = 0$, apply it to the 15 values on $C_0 \setminus \{0\}$. Their means are exactly (7) by Lemma 4.1. Take the resulting observed endpoints α_u, β_u. The range, the equality case, and the residues give conditions 1–3. Condition 4 holds because $n = \lambda_0$ occurs on C_0 and cannot occur on another coset. The same separation gives condition 5. Adding the eight bounds, with the separate term n^2 on C_0, and applying (4) gives the claimed cost bound. □

Lemma 5.3 (Finite inequality)**.** *For every case (6), every admissible bracket assignment has cost strictly greater than $128n$.*

Section 6 proves this finite assertion using exact certificates. Cases with no admissible assignment satisfy it vacuously. The parameter range includes $n = 0$, $B = \varnothing$, $t = 0$, and all other boundary values; all of them are included in the finite check.

Proof of the upper bound in Theorem 1.1. If a pairwise strongly cospectral set has more than four vertices, Lemmas 3.1 and 3.3 put its connection set in the position required by Lemma 5.2. That lemma produces an admissible assignment of cost at most $128n$, contradicting Lemma 5.3. □

6 Exact certificates for the finite inequality

We prove a lower bound that reduces Lemma 5.3 to small, independent integer calculations. For a fixed case, let J_u be the set of all brackets for coset u satisfying conditions 1–4 of Definition 5.1. At this stage disjointness between cosets is not imposed. For $j = (\alpha, \beta) \in J_u$, let $c_u(j)$ be its cost from (8) or (9).

Lemma 6.1 (Penalty bound)**.** *Assign nonnegative rational weights w_v to integers $v \in [-n, n]$. For a bracket j, let $w(j)$ be the sum of the weights on its distinct endpoints, counting a repeated endpoint once. If every J_u is nonempty, every admissible assignment has cost at least*

$$L(w) = \sum_{u=0}^{7} \min_{j \in J_u} (c_u(j) + w(j)) - \sum_{v=-n}^{n} w_v. \tag{10}$$

Proof. For the chosen bracket j_u of each coset,

$$c_u(j_u) \geq \min_{j \in J_u} (c_u(j) + w(j)) - w(j_u).$$

By disjointness, any integer is an endpoint in at most one chosen bracket. Nonnegativity therefore gives $\sum_u w(j_u) \leq \sum_v w_v$. Summing the preceding inequalities proves (10). □

This is weak duality for the linear programming relaxation in which one bracket is chosen per coset and each endpoint value has capacity one. No linear programming solver is part of the verification: nonnegative weights with $L(w) > 128n$ suffice. If some J_u is empty, there can be no admissible assignment and no weights are needed.

6.1 The finite certificate assertion

We now specify the finite arithmetic assertion used to finish the proof. There are

$$8 \sum_{B \subseteq \{1,\ldots,7\}} (128 - |B|) = 8(128 \cdot 128 - 7 \cdot 64) = 127{,}488$$

cases. A case can share its certificate with any case having the same *case key*

$$(n, m, \text{the sorted list of } (\mu_u, r_u) \text{ for } 1 \leq u \leq 7)\,. \tag{11}$$

These data determine J_0 and its cost, and determine the other seven bracket lists and costs up to permutation. Such a permutation preserves disjointness and total cost. Forming (11) for every case (6) gives 3,984 distinct keys.

A *certificate entry* consists of a case key together with either an assertion that a bracket list is empty or nonnegative integer weights W_v representing $w_v = W_v/15000$. Unlisted weights are zero. For a weight entry, the required inequality is

$$\sum_{u=0}^{7} \min_{(\alpha,\beta) \in J_u} (15000 c_u(\alpha, \beta) + W_\alpha + \mathbf{1}_{\beta \neq \alpha} W_\beta) - \sum_v W_v > 1{,}920{,}000n. \tag{12}$$

Here $\mathbf{1}_{\beta \neq \alpha}$ is one when the endpoints differ and zero when they agree. All operations are integer operations. The sets J_u are formed by ranging α, β over the integers in $[-n, n]$ and applying conditions 1–4 literally; comparisons with M_0 are made after multiplication by 15.

The finite assertion established by the supplied data is that every entry has either a genuinely empty J_u or nonnegative weights satisfying (12), and that the entries cover all cases. To *cover* means that every case's key occurs among the entries with the correct coset data. Both parts of this assertion are checked exactly, in Lean and by the separate integer checker described in Section 8. The data are supplied in `data/d7k3.cert` in the accompanying bundle.

6.2 A worked certificate

The case $n = 12$, $B = \{1, 2\}$, $t = 4$ illustrates how the weights exclude an assignment. In unscaled cost units, its certificate has the nonzero weights

v	-8	-6	-4	-2	0	2	4	6	8
w_v	256	384	896	1024	1984	1024	1216	384	704

The integer weights in the data file are 15000 times these values. In coset order $u = 0, \dots, 7$, the means and residues are

$$(M_u) = (4/3, 0, 0, -2, 2, 0, 0, -2), \qquad (r_u) = (0, 0, 0, 0, 2, 2, 2, 2).$$

Enumerating the permitted pairs gives

$$\big(\min_{j \in J_u}(c_u(j) + w(j))\big)_{u=0}^{7} = (1920, 1984, 1984, 1984, 1088, 1344, 1344, 1088),$$
$$\sum_v w_v = 7872.$$

For example, a minimizing pair on the zero coset is $(-12, 4)$. Its cost is

$$12^2 + (-12 + 4)(32 - 12) - 15(-12)(4) = 704,$$

so its adjusted cost is $704 + w_{-12} + w_4 = 704 + 0 + 1216 = 1920$. Minimizing pairs on the other cosets, in order, are $(-12, 0), (-12, 0), (-8, 8), (-10, 2), (-10, 2), (-10, 2), (-10, -2)$. Thus (10) gives

$$L(w) = 12736 - 7872 = 4864 > 1536 = 128 \cdot 12.$$

Without endpoint penalties, the sum of the eight individual minimum costs is only 608. The disjointness requirement, expressed through the penalties, is essential to this certificate.

Proof of Lemma 5.3. Each case is covered by a checked entry. An empty bracket list excludes an admissible assignment. Otherwise, the checked nonnegative weights satisfy (12), which is $15000L(w) > 15000 \cdot 128n$. Lemma 6.1 therefore puts every admissible assignment's cost strictly above $128n$. □

7 The connected example attaining four

For completeness, we describe the instance of Árnadóttir–Godsil's odd-dimensional construction [1, Theorem 10.2] that proves attainment. With the standard basis $e_1, \dots, e_7$, let

$$S = \{e_i : 1 \le i \le 7\} \cup \{e_1 + e_2, e_1 + e_3, e_2 + e_3\}$$
$$\cup \{e_i + e_j : 4 \le i < j \le 7\} \cup \{e_1 + e_2 + e_3 + e_i : 4 \le i \le 7\}.$$

These four disjoint parts have sizes $7, 3, 6, 4$, so the graph is 20-regular. The connection set contains a basis, so the graph is connected. In the integer encoding above it is

$$S = \{1, 2, 3, 4, 5, 6, 8, 15, 16, 23, 24, 32, 39, 40, 48, 64, 71, 72, 80, 96\}.$$

For this set, the character criterion gives

$$K(S) = \{0, 7, 120, 127\} = \langle e_1 + e_2 + e_3,\ e_4 + e_5 + e_6 + e_7 \rangle. \tag{13}$$

Eigenvalue θ	Multiplicity	$g = 7$	$g = 120$	$g = 127$
20	1	+	+	+
12	3	+	+	+
10	4	+	−	−
6	8	−	+	−
4	7	+	+	+
2	16	+	−	−
0	32	−	−	+
−2	24	−	+	−
−4	21	+	+	+
−6	12	+	−	−

Table 1: Character signs on the eigenvalue fibers of the connected example. Each sign is constant over the entire indicated fiber.

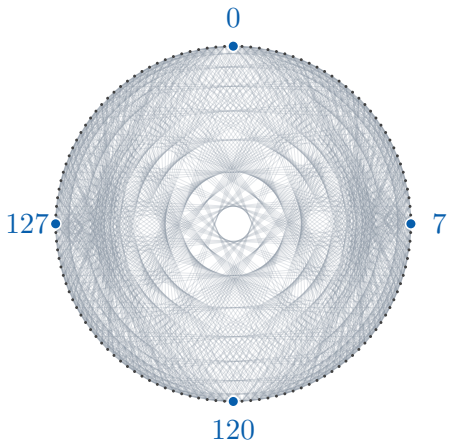


Figure 1: The connected 20-regular cubelike graph of Árnadóttir and Godsil on 128 vertices. All edges are shown in gray; the four vertices $0, 7, 120, 127$ of one strongly cospectral class are highlighted. The circular layout places these four vertices at the cardinal points to show the class within the full graph. The drawing is generated from the displayed connection set.

Figure 1 shows this connected example and highlights the four vertices in (13). Its character eigenvalue fibers provide a short explicit check of the four claimed vertices: Table 1 records the sign of $\chi_a(g)$ throughout each fiber, for the three nonzero g in (13). The sign at $g = 0$ is always positive.

For each displayed g, the value $\chi_a(g)$ is therefore constant as a ranges over an eigenvalue fiber, as required by Lemma 3.1. The upper bound already proved excludes any further vertices from this class. This establishes attainment and completes the proof of Theorem 1.1.

8 Computation and formalization

This section specifies which parts are machine-checked and what the checks assume. The mathematical upper bound is formalized from the adjacency matrix and spectral definition of strong cospectrality; the attaining example is a separate exact computation. The complete accompanying bundle contains the paper, the Lean project, the certificate data and the exact checkers. All paths in this paper are relative to that bundle: the formal definitions and theorem are

in `lean/CubelikeSC/Statement.lean` and `lean/CubelikeSC/Proof.lean`, the certificates are in `data/d7k3.cert`, and the arithmetic check is `checks/check.py`. The bundle's `README.md` gives the commands for checking it as a standalone directory.

8.1 The formal upper-bound statement

The formal definitions identify the machine-checked statement with Theorem 1.1's upper bound. The project uses Lean 4 and mathlib, with versions fixed by its Lake configuration. Its definitions are reproduced below, with line breaks for readability:

```
abbrev V := Fin 7 → ZMod 2

def adj (S : Finset V) : Matrix V V ℝ :=
  fun x y => if x - y ∈ S then 1 else 0

def StronglyCospectral (S : Finset V) (u v : V) : Prop :=
  ∀ θ : ℝ,
    Module.End.HasEigenvalue (Matrix.toEuclideanLin (adj S)) θ →
    let E := Module.End.eigenspace (Matrix.toEuclideanLin (adj S)) θ
    E.starProjection (EuclideanSpace.single u 1) =
      E.starProjection (EuclideanSpace.single v 1) ∨
    E.starProjection (EuclideanSpace.single u 1) =
      -E.starProjection (EuclideanSpace.single v 1)

def EndpointSC : Prop :=
  ∀ S : Finset V, (0 : V) ∉ S → ∀ P : Finset V,
    (∀ u ∈ P, ∀ v ∈ P, StronglyCospectral S u v) → P.card ≤ 4
```

The adjacency convention $x-y \in S$ agrees with $x+y \in S$ over $\mathbb{F}_2$. The expression `E.starProjection` is the orthogonal projection onto the real eigenspace `E`; the disjunction permits either sign separately for each eigenvalue. The name `EndpointSC` denotes the final universal upper-bound assertion. Its proof is declared as

```
theorem endpoint_sc : EndpointSC
```

The definition of `EndpointSC` is the universal statement

$$\forall S \subseteq V, \quad 0 \notin S \implies \forall P \subseteq V, \quad (\forall u, v \in P,\ u, v \text{ are strongly cospectral}) \implies |P| \leq 4.$$

It quantifies over every connection set excluding zero and every pairwise strongly cospectral set, including disconnected graphs.

The formalization proves the direction of Lemma 3.1 needed for the upper bound: strong cospectrality of u, v implies $v - u \in K(S)$. It proves the subspace and parity facts, the moment identities, bracket extraction, the penalty bound, the finite certificate checks, and their case cover. Instead of changing basis, it selects three independent vectors of $K(S)$ whose span contains σ, and indexes character cosets by their three dot products. This is the same reduction in a coordinate-independent form.

All certificate calculations are evaluated by Lean's kernel. No axiom asserts the finite inequality, and `native_decide` is not used. The final theorem depends only on the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The build fails if the theorem's axiom dependencies differ from this list. From `lean/`, the standard check is

```
lake exe cache get
lake build
```

The trusted foundation for that check is Lean's kernel and the definitions and library results used by the proof. The formal theorem's identification with the graph-theoretic statement is given by the definitions just described, rather than by a hypothesis about character eigenvalue fibers.

8.2 Independent arithmetic checks

The separate integer checker verifies the certificates directly for the original cases, and also checks the rejection of deliberately damaged data. The resulting certificate counts are summarized in Table 2. The 58 empty-list entries have $n = 0, 1, 2, 3, 4, 5, 6$, with respectively $2, 5, 9, 15, 16, 9, 2$ keys.

Certificate disposition	Number of case keys
A coset has no allowed bracket	58
Nonnegative weights satisfy (12)	3,926
Total	3,984

Table 2: Exact certificate outcomes. Each key can represent several cases; these are counts of keys, not counts of connection sets or cases.

Together they represent 1,628 original cases. For each entry, forming the relevant J_u verifies that no integer pair satisfies conditions 1–4. For example, at $n = 0$ the only possible endpoint is zero, which condition 4 forbids on a nonzero coset. A separate exact Python checker forms each case's own eight bracket lists and checks the matching certificate on those lists, without assuming the key-sharing argument.

8.3 Certificate generation and the lower example

We distinguish the untrusted generation of certificates from their exact verification, and record the separate check of attainment. The certificate weights were obtained from a floating-point linear programming dual and converted to rational numbers. This generation step is not trusted: only the inequalities on the supplied integer weights are used. The separate Python certificate checker uses arbitrary-precision integers. Its role is an additional implementation of the certificate calculation, not an additional mathematical assumption in the Lean theorem. To test rejection, the supplied checks deliberately remove an entry, falsely assert that a bracket list is empty, weaken or zero weights, make a weight negative, and duplicate an entry. Every altered file is rejected. Lean also proves that selected entries fail their check when weights are removed or halved, and that deleting a required entry leaves a case without a certificate. The command `python3 checks/check.py` performs the exact Python checks and these rejection tests; `checks/README.md` explains them.

Earlier computational work is recorded in Árnadóttir and Godsil's Appendix A, which gives six connection sets on 32 vertices; their complements give the other six graphs. Appendix B provides Sage code. Their acknowledgments credit Gordon Royle's cubelike graph lists.

The converse of Lemma 3.1 and attainment are not part of the Lean development. For the displayed construction, exact character sums check (13); a separate computation checks strong cospectrality using spectral projection columns with rational entries. The construction and the written character argument above give the lower bound four. Neither the upper theorem nor this lower example relies on a numerical eigensolver.

Figure 1 and its SVG version in `data/figure.svg` come from the same graph data. The optional illustration script `checks/figure.py` checks the graph's vertex count, edge count, degree, connectivity and highlighted class, and reproduces the figure and the spectral-sign table. It is not needed for the upper-bound proof.

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References

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